Topics in Error-Correcting Codes

Notes by Rohan Goyal

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Lecture 14: Optimal List-Decoding
of Subspace-Design Codes

Independent Codewords: Computing the Potential

We revisit Lecture 12's independent-codeword argument using potentials. Let $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ be additive, with all dimensions over $\mathbb{F}_q$.

Recall that an r-LCL profile $\mathcal{V} = (V_1, \dots, V_n)$ consists of maps $V_i : \mathbb{F}_q^r \to \mathbb{F}_q^r$. For $W \leq \mathbb{F}_q^r$, its potential is

$$\Phi_W(\mathcal{V}, \alpha) = \alpha \dim W - \frac{1}{n} \sum_{i=1}^{n} \big(\dim W - \dim(W \cap \ker V_i)\big).$$

In particular,

$$\Phi_{\mathbb{F}_q^r}(\mathcal{V}, \alpha) = \alpha r - \frac{1}{n} \sum_i \big(r - \dim \ker(V_i)\big).$$

Each $r - \dim \ker(V_i)$ counts the independent equations at coordinate i.

For $m \geq 2$, let

$$\delta < \frac{m-1}{m}\big(1 - \tau(m)\big).$$

Suppose $c_0, \dots, c_{m-1}$ are affinely independent and $\Delta(c_j, y) \leq \delta$ for every j. Without loss of generality, $c_0 = 0$. Then $c_1, \dots, c_{m-1}$ are linearly independent. For each coordinate, let

$$e_i = \{j \in \{0, \dots, m-1\} : c_{j,i} = y_i\}.$$

Define $V_i : \mathbb{F}_q^{m-1} \to \mathbb{F}_q^{m-1}$ by specifying its kernel:

$$\boxed{\ker(V_i) = \{x \in \mathbb{F}_q^{m-1} : x_j = x_\ell \text{ for all } j, \ell \in e_i\},}$$

where $x_0 = 0$. Then

$$(m-1) - \dim \ker(V_i) = \max\{|e_i| - 1, 0\}.$$

The independent words $c_1, \dots, c_{m-1}$ realize the profile. Hence

$$\begin{aligned}
\sum_i \big((m-1) - \dim \ker(V_i)\big) &= \sum_i \max\{|e_i| - 1, 0\} \\
&\geq \sum_i |e_i| - n \\
&\geq n\big(m(1-\delta) - 1\big) \\
&> n(m-1)\tau(m).
\end{aligned}$$

For a τ-subspace-design code, universal noncontainment and the monotonicity of τ give

$$0 \le \Phi_{\mathbb{F}_q^{m-1}}(\mathcal{V}, \tau(m-1)) \le \Phi_{\mathbb{F}_q^{m-1}}(\mathcal{V}, \tau(m)) < 0,$$

a contradiction.

This controls affine independence in a nearby list. To bound the number of distinct codewords, we must also handle dependent lists.

Local Profiles for Dependent Codewords

A list can contain

$$c_1, \qquad c_2, \qquad 2c_1 - c_2,$$

whose agreement equations can be dependent.

A three-word example

Suppose c_1, c_2 are independent and the field has odd characteristic, so these three words are distinct. If all three agree at coordinate i, the equations on the symbols of c_1, c_2 are

$$x_1 = x_2 = 2x_1 - x_2.$$

Thus

$$\ker(V_i) = \{(x_1, x_2) : x_1 = x_2 = 2x_1 - x_2\} = \{(x, x) : x \in \mathbb{F}_q\}.$$

This kernel has dimension one, so there is only one independent equation, although $|e_i| - 1 = 2$. For this three-word list, any pair gives the same equation $x_1 = x_2$. Therefore the full definition is

$$\ker(V_i) = \begin{cases} \mathbb{F}_q^2, & |e_i| \le 1, \\ \{(x, x) : x \in \mathbb{F}_q\}, & |e_i| \ge 2. \end{cases}$$

The general construction below works over every finite field.

The profile of an arbitrary list

Now take distinct $c_0, \dots, c_{m-1}$ with $\Delta(c_j, y) \le \delta$, without assuming affine independence. Without loss of generality, $c_0 = 0$ and $c_1, \dots, c_r$ form a basis for the span of the list. Write

$$c_j = \sum_{h=1}^{r} a_{j,h} c_h, \qquad a_j \in \mathbb{F}_q^r, \qquad a_0 = 0.$$

The vectors a_j are distinct and span $\mathbb{F}_q^r$, with $r \le m - 1$. Keep the agreement sets $e_i = \{j : c_{j,i} = y_i\}$.

Define $V_i : \mathbb{F}_q^r \to \mathbb{F}_q^r$ by

$$\boxed{\ker(V_i) = \left\{ x \in \mathbb{F}_q^r : \sum_{h=1}^{r} (a_{j,h} - a_{\ell,h}) x_h = 0 \text{ for all } j, \ell \in e_i \right\}.}$$

The independent words $c_1, \dots, c_r$ realize this profile coordinatewise. Its potential is

$$\Phi_{\mathbb{F}_q^r}(\mathcal{V}, \alpha) = \alpha r - \frac{1}{n} \sum_i \big(r - \dim \ker(V_i)\big).$$

The difficulty is now to compute $\dim \ker(V_i)$. Knowing $|e_i|$ does not suffice: some of its equality constraints may follow from the others.

Counting Constraints with a Partition

This partitioning idea is due to Yeyuan Chen and Zihan Zhang in their breakthrough paper *Explicit Folded Reed–Solomon and Multiplicity Codes Achieve Relaxed Generalized Singleton Bounds*. They proved that these codes achieve list-decoding capacity with optimal list size $O(1/\varepsilon)$.

Set

$$A_0 = \{0\}, \qquad A_h = \operatorname{span}\{c_1, \dots, c_h\} \quad (1 \le h \le r).$$

Define $P = \{P_0, \dots, P_r\}$ by

$$P_0 = \{0\}, \qquad P_h = \{j : c_j \in A_h \setminus A_{h-1}\} \quad (1 \le h \le r).$$

Every part is nonempty: $h \in P_h$. For a partition P, write

$$P(e) = |\{h : e \cap P_h \neq \varnothing\}| \quad (e \neq \varnothing), \qquad P(\varnothing) = 1.$$

Thus $P(e) - 1$ counts equalities among representatives from the parts e intersects; an empty edge contributes zero.

Lemma 1 (Independent representatives)**.** *Choose $0 \le j_1 < \cdots < j_t \le r$ and $k_\ell \in P_{j_\ell}$ for $1 \le \ell \le t$. Then $c_{k_1}, \dots, c_{k_t}$ are affinely independent:*

$$\dim \operatorname{span}\{c_{k_\ell} - c_{k_1} : 2 \le \ell \le t\} = t - 1.$$

Proof. For $2 \le \ell \le t$, we have $c_{k_\ell} - c_{k_1} \in A_{j_\ell} \setminus A_{j_\ell - 1}$, so these differences are linearly independent. □

For a nonempty e_i, select one codeword from each part it intersects. Their coefficient vectors are affinely independent, so equating their symbols gives $P(e_i) - 1$ independent equations in the definition of $\ker(V_i)$. Therefore, including the empty case,

$$r - \dim \ker(V_i) \ge P(e_i) - 1,$$

and

$$\boxed{\Phi_{\mathbb{F}_q^r}(\mathcal{V}, \alpha) \le \alpha r - \frac{1}{n} \sum_i \big(P(e_i) - 1\big).}$$

Why this does not finish the proof

The total agreement weight $\sum_i \max\{|e_i| - 1, 0\}$ may be large, while $\sum_i (P(e_i) - 1)$ is small. Many agreements may be within one part. They can overconstrain a smaller space while leaving the whole-space potential nonnegative.

For example, in odd characteristic take the list

$$c_0 = 0, \qquad c_1, \qquad c_2, \qquad c_3 = 2c_1,$$

where c_1, c_2 are independent. Suppose $e_i = \{0,1,3\}$ on $4n/5$ coordinates and $e_i = \varnothing$ on the others. Then

$$\ker(V_i) = \begin{cases} \{(x_1,x_2) : 0 = x_1 = 2x_1\} = \{0\} \times \mathbb{F}_q, & e_i = \{0,1,3\}, \\ \mathbb{F}_q^2, & e_i = \varnothing. \end{cases}$$

There is one equation at each of the $4n/5$ active coordinates. At $\alpha = 1/2$,

$$\Phi_{\mathbb{F}_q^2}(\mathcal{V}, 1/2) = 2 \cdot \frac{1}{2} - \frac{4}{5} = \frac{1}{5} > 0.$$

The parts are $\{0\}, \{1,3\}, \{2\}$, so $P(e_i) - 1 = 1$ at each active coordinate. Although $\sum_i \max\{|e_i| - 1, 0\} = 8n/5$, the partition gives only $4n/5$ independent equations.

Retain just c_0, c_1, c_3. Now the profile has locality one: its kernel is $\{0\}$ on $4n/5$ coordinates and $\mathbb{F}_q$ elsewhere. Its potential is

$$\frac{1}{2} - \frac{4}{5} = -\frac{3}{10}.$$

This is the quotient from Lecture 13. Take $W = \{0\} \times \mathbb{F}_q$. Since $W \subseteq \ker(V_i)$ for every i, $\Phi_W(\mathcal{V}, 1/2) = 1/2$. Projecting by $\pi(x_1, x_2) = x_1$ gives

$$\Phi_{\mathbb{F}_q}(\widehat{\mathcal{V}}, 1/2) = \Phi_{\mathbb{F}_q^2}(\mathcal{V}, 1/2) - \Phi_W(\mathcal{V}, 1/2) = \frac{1}{5} - \frac{1}{2} = -\frac{3}{10}.$$

The following claim gives such a subset.

Finding a WPC Subset

View the agreement sets as a hypergraph

$$H = (X, (e_i)_{i=1}^n), \qquad X = \{0, \dots, m-1\}.$$

Edges are indexed by code coordinates: repeated edges count separately, and empty edges are allowed.

WPCs were first used in this context by Omar Alrabiah, Venkatesan Guruswami, and Ray Li in *Randomly Punctured Reed–Solomon Codes Achieve List-Decoding Capacity over Linear-Sized Fields*. They proved that random puncturings of Reed–Solomon codes achieve list-decoding capacity over fields of size $O(n)$.

Definition 2 (Weak partition connectivity)**.** A hypergraph is *t-weakly partition-connected*, or *t-WPC*, if every partition P of X into nonempty parts satisfies

$$\sum_{i=1}^{n} \big(P(e_i) - 1\big) \geq t(|P| - 1).$$

Here $t \geq 0$ is real. For our partition, $|P| - 1 = r$, so WPC gives the required bound tr.

Claim 3 (A WPC subset)**.** *Suppose $|X| \geq 2$ and*

$$\sum_{i=1}^{n} \max\{|e_i| - 1, 0\} \geq t(|X| - 1).$$

Then there is a subset $S \subseteq X$, with $|S| \geq 2$, for which the induced hypergraph

$$\big(S, (e_i \cap S)_{i=1}^n\big)$$

is t-WPC.

WPC holds for every partition of the selected subset, including the one defined above.

Project to the Selected Words and Count Constraints

Fix $\alpha \geq 0$ and suppose an agreement configuration on $2 \leq |X| \leq L+1$ distinct codewords satisfies

$$t := \frac{\sum_{i=1}^{n} \max\{|e_i| - 1, 0\}}{|X| - 1} > \alpha n.$$

Claim 3 gives a t-WPC subset S with at least two vertices.

Restrict to these vertices, replacing each edge by $e_i \cap S$, and relabel. Without loss of generality, the selected list is $c_0 = 0, c_1, \ldots, c_{m-1}$, with $c_1, \ldots, c_r$ independent and spanning the list. Here $1 \leq r \leq m-1 \leq L$. As before, write $c_j = \sum_h a_{j,h} c_h$ and define

$$\ker(V_i) = \{x \in \mathbb{F}_q^r : a_j^\mathsf{T} x = a_\ell^\mathsf{T} x \text{ for all } j, \ell \in e_i\}.$$

These independent codewords realize the profile.

This is the projection from Lecture 13: the selected independent differences are fixed linear combinations of the original basis words. Applying those combinations to the symbols gives a surjective linear map. Its image of each original kernel satisfies all the selected equalities, and is therefore contained in the new $\ker(V_i)$ above.

Apply the partition above to this list. Since $|P| - 1 = r$, the partition bound and WPC give

$$\sum_i \big(r - \dim \ker(V_i)\big) \geq \sum_i \big(P(e_i) - 1\big) \geq tr.$$

Thus

$$\boxed{\Phi_{\mathbb{F}_q^r}(\mathcal{V}, \alpha) \leq \alpha r - \frac{tr}{n} = r\left(\alpha - \frac{t}{n}\right) < 0.}$$

For a τ-subspace-design code, take $\alpha = \tau(L)$. Since $r \leq L$ and τ is nondecreasing,

$$\Phi_{\mathbb{F}_q^r}(\mathcal{V}, \tau(r)) \leq \Phi_{\mathbb{F}_q^r}(\mathcal{V}, \tau(L)) < 0.$$

Universal noncontainment from Lecture 13 rules this out. We have proved that every agreement configuration on $m \leq L+1$ distinct codewords satisfies

$$\sum_i \max\{|e_i| - 1, 0\} \leq n\tau(L)(m-1).$$

This gives us the Chen-Zhang list-decoding bound.

Theorem 4 (Chen–Zhang list-decoding bound)**.** *Let $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ be a τ-subspace-design code, with τ nondecreasing. For every $L+1$ distinct codewords $c_0, \ldots, c_L$ and every received word y,*

$$\frac{1}{L+1} \sum_{j=0}^{L} \Delta(c_j, y) \geq \frac{L}{L+1}\big(1 - \tau(L)\big).$$

In particular, for every $\rho < \frac{L}{L+1}(1 - \tau(L))$, the code is (ρ, L) average-radius list-decodable, and hence (ρ, L) list-decodable.

Recall the generalized Singleton bound from Lecture 12: a rate-R code with list size L cannot asymptotically have radius larger than $\frac{L}{L+1}(1 - R)$. Thus when $\tau(L)$ is close to R, the theorem approaches the optimal radius for that list size.